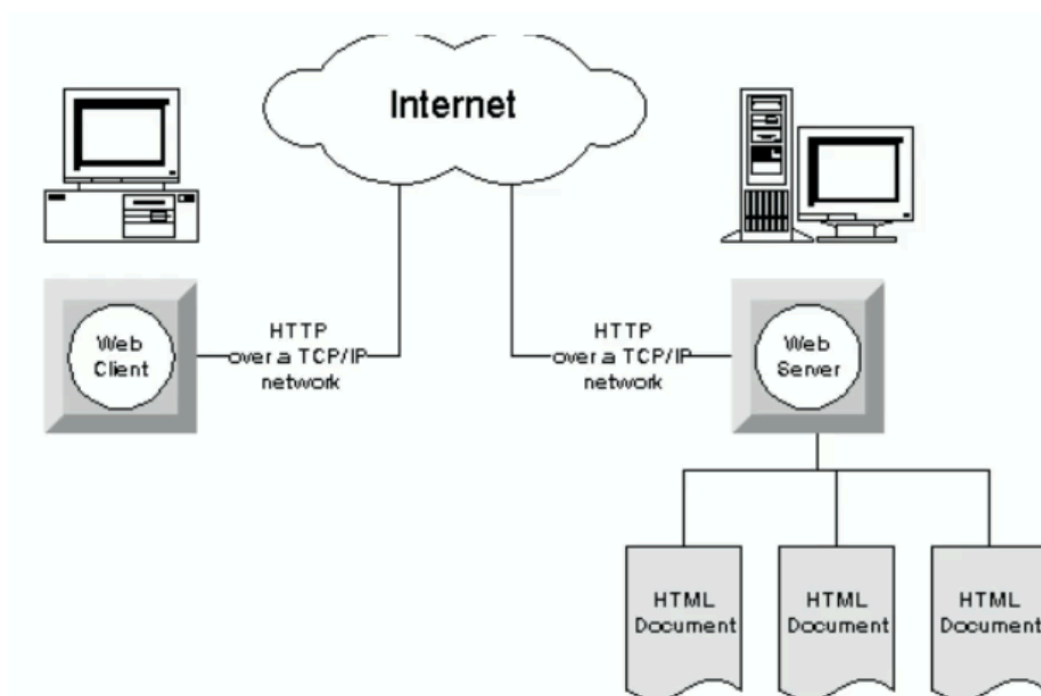


Section 5: Web Concepts in System Design

Introduction to Web Concepts

In this lecture, we will explore the web concepts that sit behind every modern application and learn how they improve scalability, security, performance & architectural decision making in real-world system design.

Why Learn Web Concepts?



Web concepts are the fundamental part of system design. Nowadays, almost every large-scale application, whether it's an e-commerce platform, social network or cloud service, is built on top of web technologies. As architects and engineers, we cannot design effective systems without understanding how these web components interact. What makes web concepts especially important is their impact on scalability, security and performance. As systems grow from hundreds to millions of users, decisions around state management, communication patterns, caching, authentication and data exchange start to have a significant architectural impact. The difference between a system that scales smoothly and the one that struggles under load often comes down to understanding these foundational web principles. These concepts are also highly relevant in system design interviews. Interviewers frequently explore topics such as session management, cookies, caching, cores, authentication and browser-server interaction. Not because they are web-specific trivia, but because they reveal how well you understand the behavior of real-world systems.

Throughout this lecture, we will build a solid understanding of these web fundamentals and more importantly, learn how they influence architectural decisions in modern scalable systems. The goal of this section is not just to understand how the web concepts work, but to think like an architect who can leverage these concepts to design better systems .

Section Agenda

- Lecture 1: Web Sessions: Managing State in Web Applications
- Lecture 2: Serialization (Data Exchange & Storage Formats)
- Lecture 3: CORS (Cross-Origin Resource Sharing) & Web Security
- Lecture 4: Summary & Practical Applications

By the end of this section, you will not only understand these web fundamentals, but also know how they influence the design of a scalable, secure, and production-ready system.